New Uzbekistan University

Sustainable Aviation Technologies
Current Status and Future Prospects

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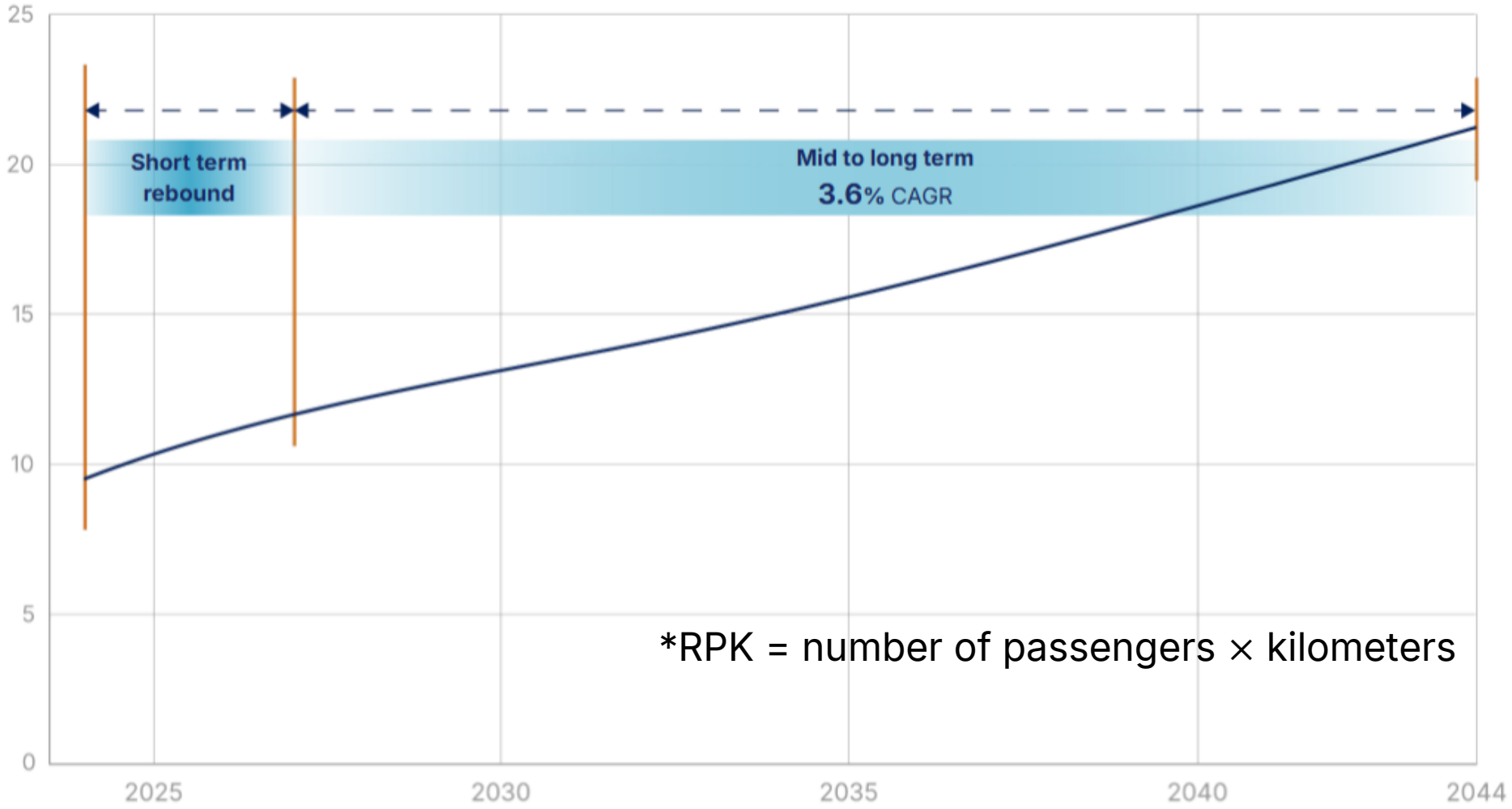
March 26, 2026

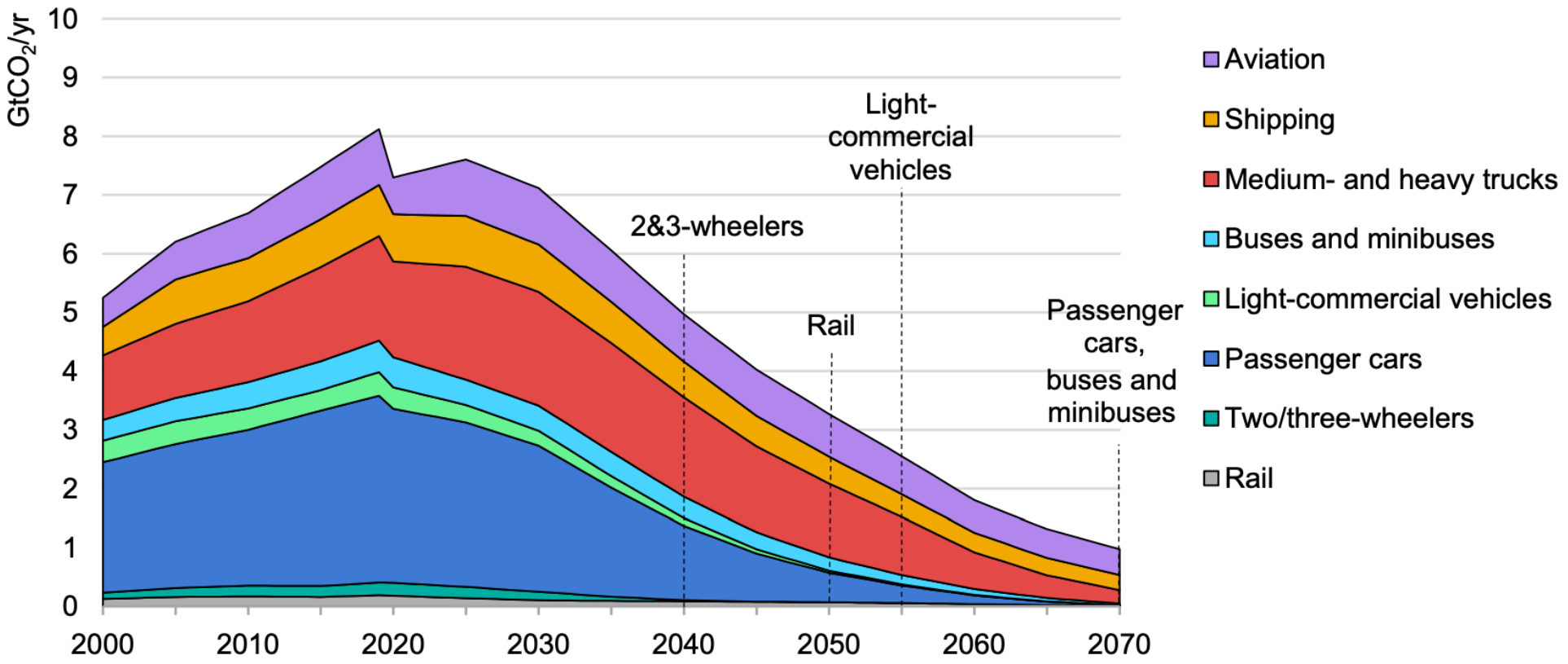
World air traffic (trillion RPK)*



*RPK = number of passengers × kilometers

Source: IATA, Airbus GMF 2025
*Revenue Passenger Kilometers, includes scheduled and non-scheduled flights

Figure 3.16 Global CO_2 emissions in transport by mode in the Sustainable Development Scenario, 2000-70



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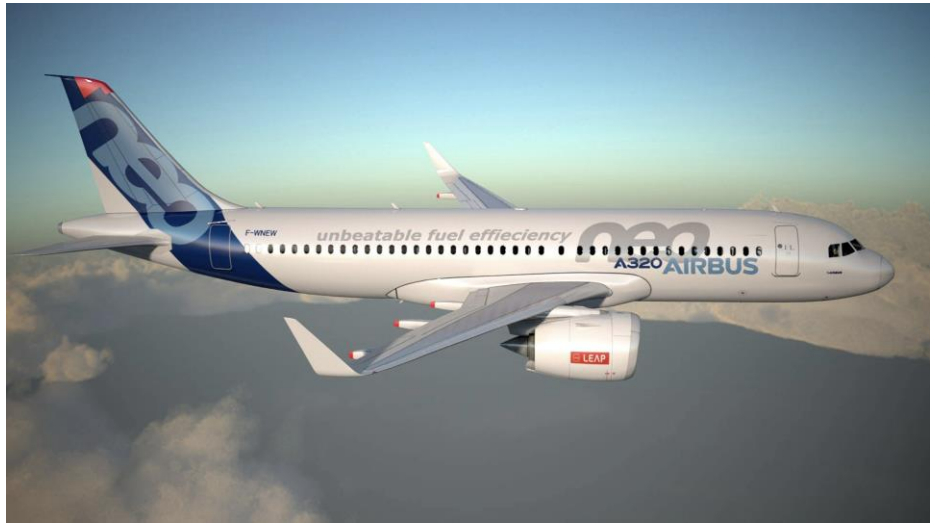
Notes: Dotted lines indicate the year in which various transport modes have largely stopped consuming fossil fuels and hence no longer contribute to direct emissions of CO_2 from fossil fuel combustion. Residual emissions in transport are compensated by negative emissions technologies, such as BECCS and DAC, in the power and other energy transformation sectors.

https://ourworldindata.org/co2-emissions-from-transport

Modern Aircraft Are Already Very Efficient



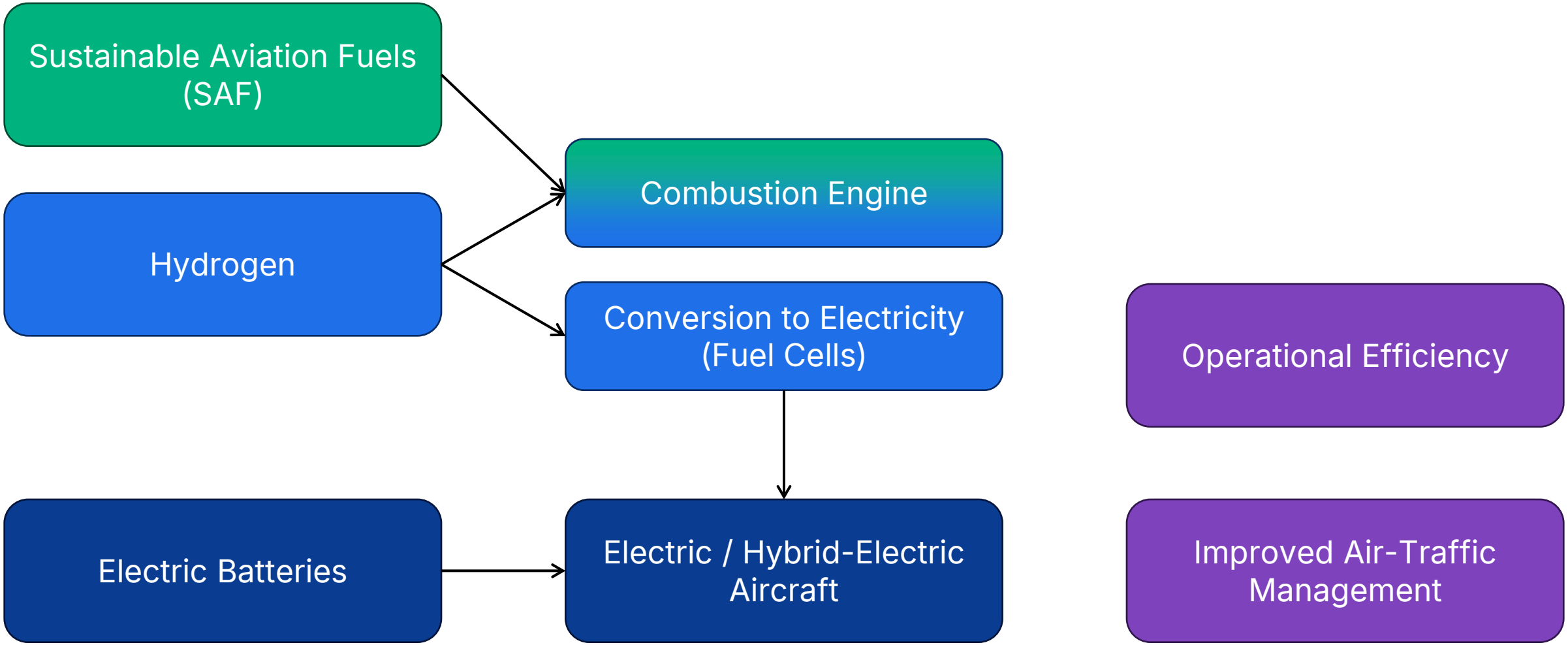
2 – 3
liters / (100 km * passenger)

- **Why aircraft are so efficient?**
  - Very high passenger density
  - Optimized engines
  - Operation near optimal condition (cruise)
- Further improvement is very difficult

Driver Only (1 person)
6 – 10
liters / (100 km * passenger)

4 People
1.5 – 2.5
liters / (100 km * passenger)

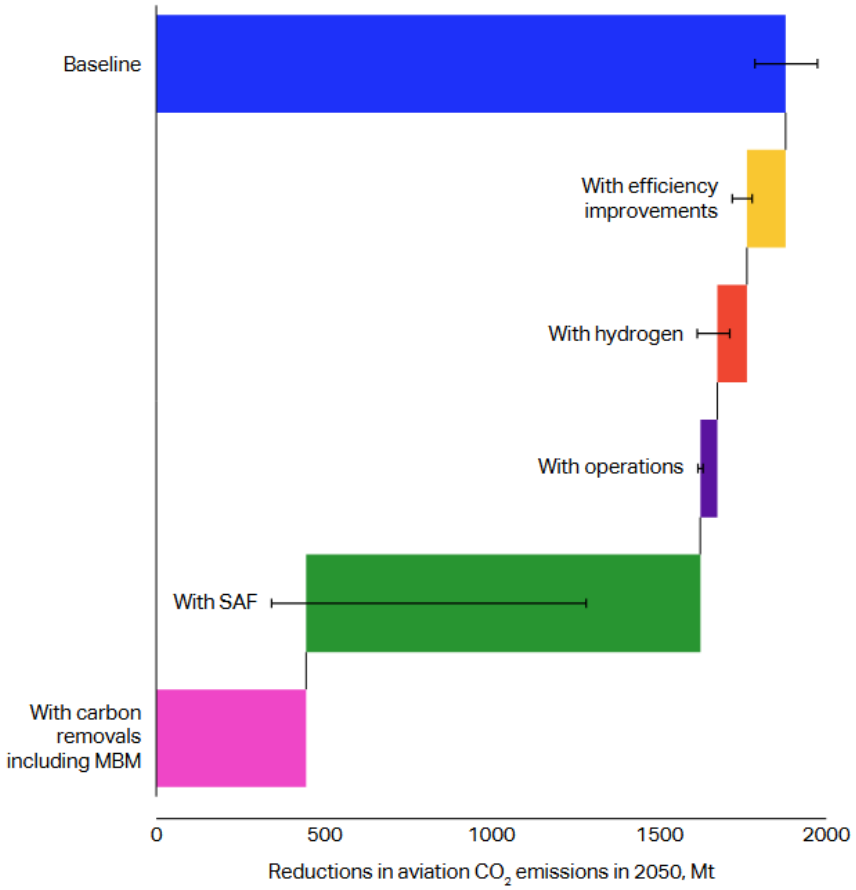
Aviation Decarbonization Strategies



Sustainable Aviation Fuel (SAF)

- A drop-in fuel compatible with existing aircraft and infrastructure
- Produced from:
  - Biomass (waste oils, residues)
  - Synthetic fuels (Power-to-Liquid)
- Up to ~80% lifecycle CO_2 reduction vs fossil jet fuel
- Key solution for long-haul aviation
- Challenges
  - Very low current share (<1%)
  - High-cost vs conventional fuel
  - Dependence on feedstock and energy

Chart 2: Reduction in aviation CO_2 emissions in 2050 achieved through the different levers of action. The solid bar indicates the central case and the black lines indicate maximum and minimum reductions based on the scenarios modeled.

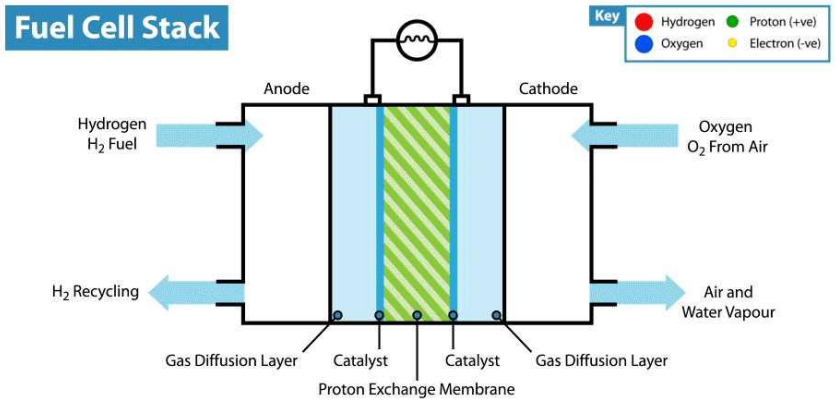


Source: IATA Sustainability and Economics, ICAO LTAG SAF availability scenarios

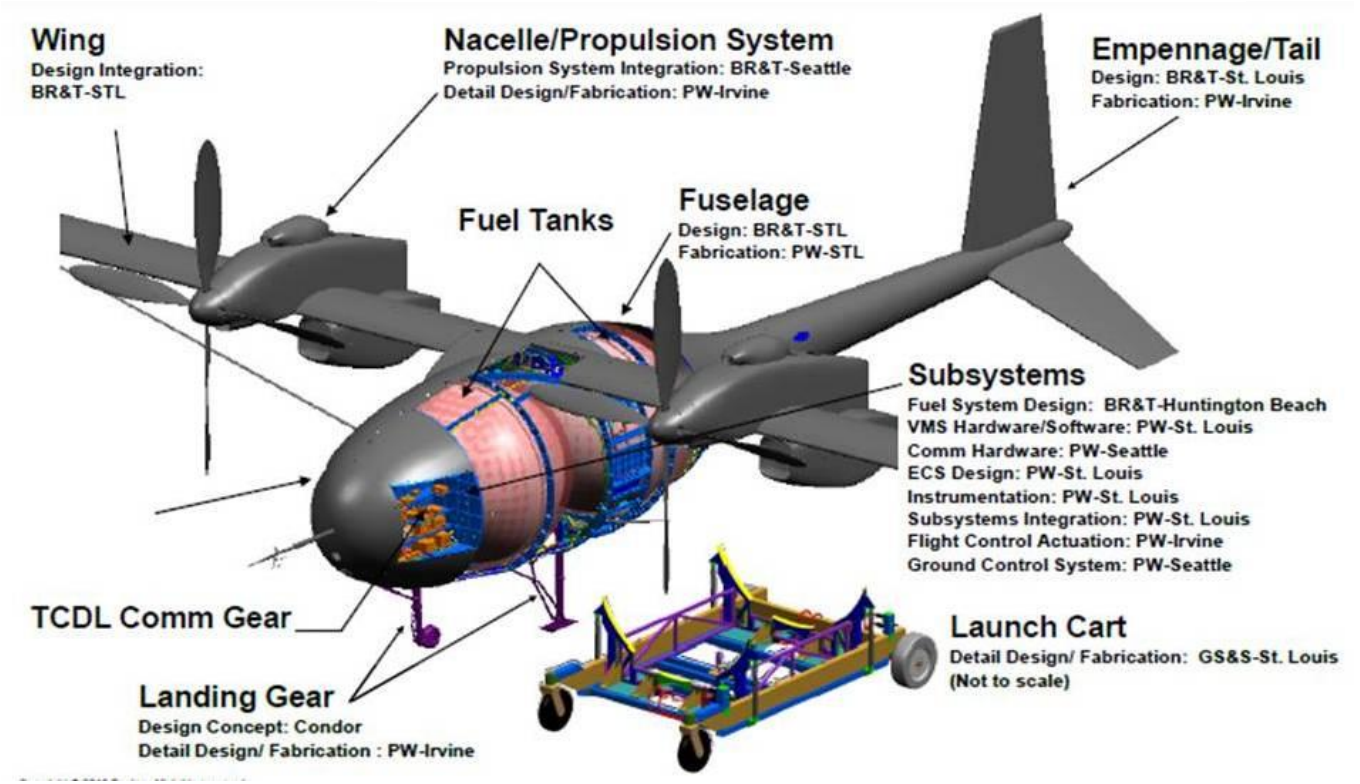
https://www.energy.gov/eere/bioenergy/sustainable-aviation-fuels
IATA Report "Energy and New Fuels Infrastructure Net Zero Roadmap"

Hydrogen

- Two main paths
  - **Combustion** in modified gas turbines
    - No CO_2, but still produces NOx
  - **Conversion** to electricity via fuel cells
    - Zero CO_2 and low emissions



- Characteristics
  - Very high energy per mass
  - Very low energy per volume



- Challenges
  - New aircraft design
  - New infrastructure
  - High-pressure or cryogenic storage

Recent Revolutions in Electric Transportation

- Electric cars started by Tesla

- Drones started by DJI

- AAM – an umbrella term for next generation aircraft
- Remotely piloted, autonomous, electric/hybrid, vertical takeoff and landing

Advanced Air Mobility

Unmanned Aerial System (UAS)

Urban Air Mobility (UAM)

Regional Air Mobility (RAM)

https://en.wikipedia.org/wiki/Advanced_Air_Mobility

Urban Air Mobility – top layer for urban mobility

- Small electric vertical takeoff and landing highly automated aircraft
- Transportation of passengers and cargo
- Low altitude within urban and suburban areas

UAM Aircraft – larger, safer and more efficient version of a drone

VOLOCOPTER

- 1 pilot, 1 passenger
- 27 km range
- Speed 70 km/h
- Gross weight 450 kg

wisk

- Speed 180 km/h
- All electric
- 2-seater
- Autonomous
- 100 km range with reserves

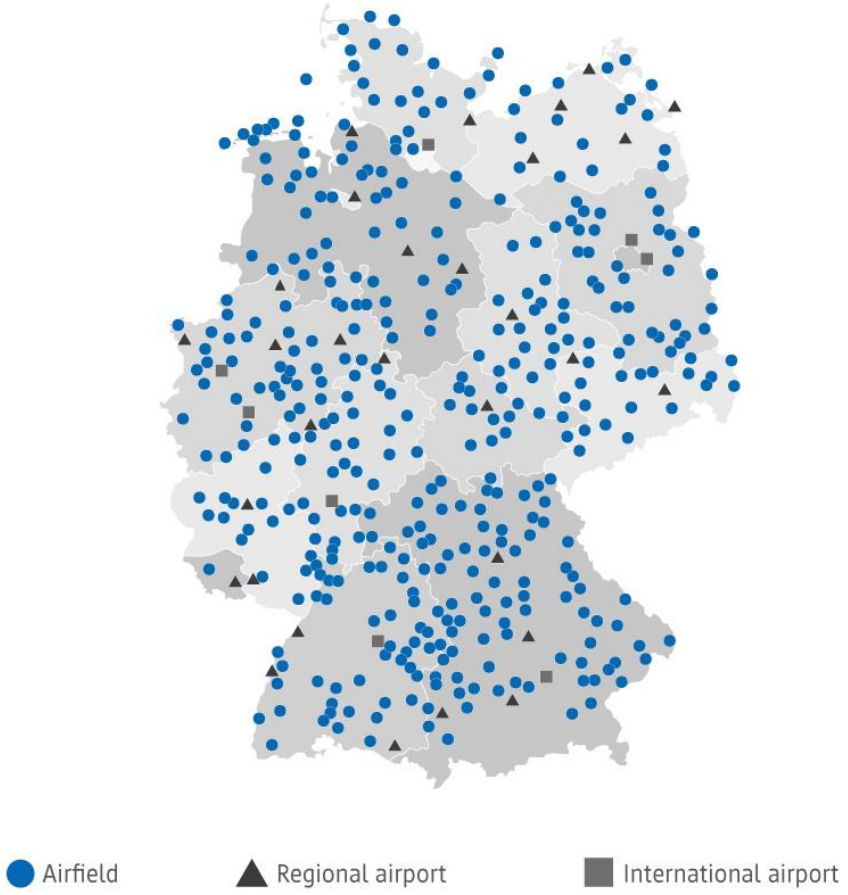
Joby

- All electric
- 240 km range
- 320 km maximum speed
- Piloted
- 1 pilot and 4 passengers

* Reference specs from open sources. Official data is very limited

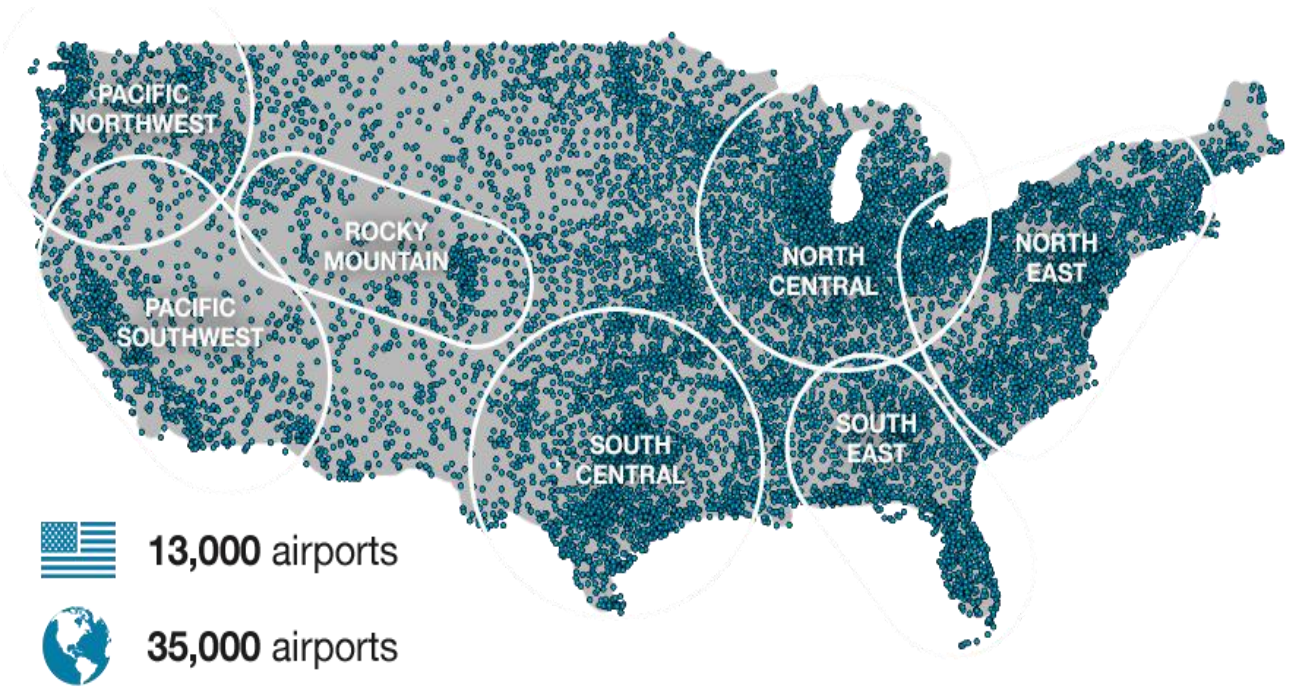
Regional Air Mobility

- An emerging market that is expected to raise in next 10-15 years
- Future short-range electric RAM aircraft will probably have the following characteristics
  - Short haul flight (100-400 km)
  - Fully electric or hydrogen fuel cell powered
  - Operations from small airfields
  - Part 23 certified



https://e-sat.de/en/regional-air-mobility/

Use Cases for Regional Air Mobility



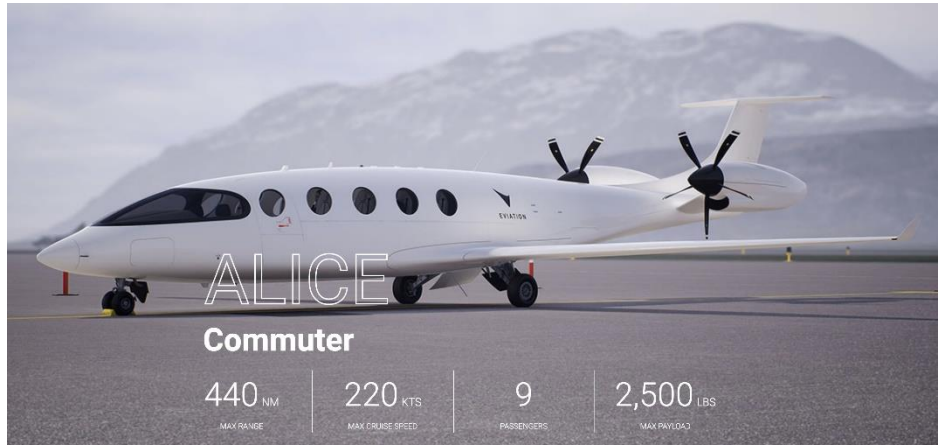
https://zunum.aero/technology/

- Use existing airports and airfields
- Stand alone travel (100-400 km)
  - Inter-city
  - Inter-island
  - City-to-rural
  - City-to-beach
- Commercial travel
  - Feeders to major hubs (Incheon, Frankfurt, JFK)
  - Route opener (new thin routes)
  - Potentially replace conventional aircraft with more energy efficient service

A Thin Route in Uzbekistan

- Antonov AN-2 operates daily between Fergana and the Sokh exclave
- 75 km route
- Electric regional aircraft is a perfect replacement

Projects by Industry Leaders and Startups



Eviation Alice
The first commercial electric RAM that started flight tests in 2022

Cessna Caravan upgraded with MagniX electric powertrain

NASA Maxwell X-57
Distributed electric propulsion (DEP) demonstrator
Tecnam P2002 with newly designed wing

What's Next for Aviation

- Aviation is a small, but rapidly growing source of emissions
- Modern aircraft are already highly optimized
- Main decarbonization paths
  - Short term: sustainable aviation fuel
  - Medium term: regional electric aircraft
  - Long term: hydrogen, operational efficiency

https://www.youtube.com/watch?v=EKG-6rxXAXE